

North Carolina A&T State University

Department of Mechanical Engineering

MEEN 424

Vibrations & Control

Session 3 Lecture Notes

Modeling in the Frequency Domain

Chapter 2 — Laplace Transform Review

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Reference: Nise, *Control Systems Engineering*, Ch. 2

These notes are prepared for students enrolled in MEEN 424.
They supplement, but do not replace, attendance and active participation.

Contents

1 Chapter 2: Modeling in the Frequency Domain	2
1.1 Motivation	2
2 Section 2.2: Laplace Transform Review	2
2.1 Definition	2
3 Calculus Review: Rules for Derivatives and Integrals	3
3.1 Chain Rule	3
3.2 Product Rule	3
3.3 Basic Integral Rules	4
3.4 Substitution Rule (U-Substitution)	4
4 Computing the Laplace Transform from the Definition	4
4.1 Example 2.1 — Laplace Transform of $Ae^{-at}u(t)$	4
4.2 Computing $\mathcal{L}\{e^{at}\}$	5
5 Properties of the Laplace Transform	5
5.1 Linearity	6
5.2 Differentiation in Time	6
5.3 Multiplication by t (Frequency Differentiation)	6
5.4 Integration in Time	6
5.5 Frequency Shift (First Shift Theorem)	7
6 Table of Laplace Transforms	7
7 Worked Examples	8
7.1 Using Linearity Directly from the Table	8
7.2 Using Entry #30: Multiplication by t^n	9
7.3 Using Entry #24: Scaling in Time	10
8 Key Formulas Summary	11
Glossary of Key Terms	12

1. Chapter 2: Modeling in the Frequency Domain

1.1. Motivation

A system represented by a differential equation is difficult to model as a block diagram. The Laplace transform allows us to represent the input, output, and system as **separate entities** whose interrelationship is simply **algebraic** rather than differential.

Key advantages of the Laplace transform approach:

- Converts differential equations into algebraic equations
- Allows block diagram representation of systems
- Handles non-zero initial conditions naturally
- Provides the transfer function concept directly
- Handles functions discontinuous at $t = 0$ using the lower limit 0^-

2. Section 2.2: Laplace Transform Review

2.1. Definition

The **Laplace transform** of a time function $f(t)$ is defined as:

$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt \quad (2.1)$$

where $s = \sigma + j\omega$ is a complex variable.

Notes on the definition:

- $s = \sigma + j\omega$ is a complex variable with real part σ and imaginary part ω .
- The lower limit 0^- means that even if $f(t)$ is discontinuous at $t = 0$, we can start the integration *prior* to the discontinuity as long as the integral converges.
- This property has distinct advantages when the initial conditions are discontinuous at $t = 0$.
- Using differential equations, we must solve for initial conditions *after* the discontinuity. Using the Laplace transform, we need only know initial conditions *before* the discontinuity.
- The result $F(s)$ is called the **Laplace transform** of $f(t)$.

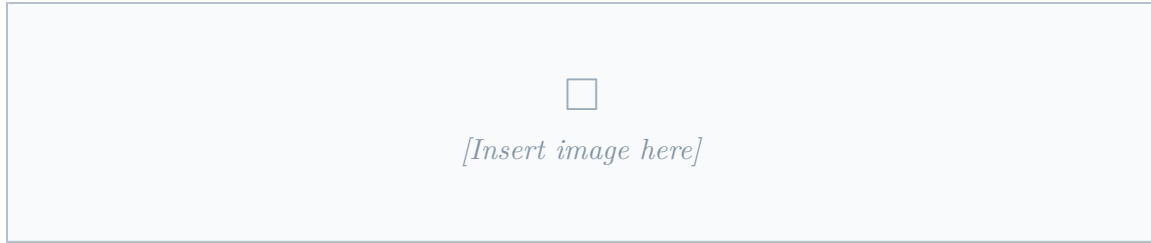


Figure 1: The Laplace transform definition and the relationship between $f(t)$ and $F(s)$

3. Calculus Review: Rules for Derivatives and Integrals

Before computing Laplace transforms, we review the key calculus rules used in the derivations.

3.1. Chain Rule

If $y = F(u)$ and $u = g(x)$, then:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Examples:

1. $f(x) = e^x \Rightarrow f'(x) = e^x$

2. $f(x) = e^{2x}$

Let $y = e^u$, $u = 2x \Rightarrow \frac{dy}{du} = e^u$, $\frac{du}{dx} = 2$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot 2 = 2e^{2x}$$

3.2. Product Rule

$$\frac{d}{dx} [f(x) \cdot g(x)] = f'(x) g(x) + f(x) g'(x)$$

Example:

3. $f(x) = xe^x$

$$(xe^x)' = (x)'(e^x) + (x)(e^x)' = e^x + xe^x$$

3.3. Basic Integral Rules

4. $\int e^x dx = e^x + C$

Since $(e^x)' = e^x$, the antiderivative is e^x .

5. $\int e^{e^x} \cdot e^x dx$

Let $u = e^x \Rightarrow du = e^x dx$:

$$\int e^u du = e^u + C = e^{e^x} + C$$

3.4. Substitution Rule (U-Substitution)

$$\int f(g(x)) g'(x) dx = \int f(u) du \quad \text{where } u = g(x)$$

Examples:

6. $\int e^{e^x} \cdot e^x dx$

Let $u = e^x \Rightarrow du = e^x dx$:

$$\int e^u du = e^u + C = e^{e^x} + C$$

7. $\int \left(1 - \frac{1}{w}\right) \cos(w - \ln w) dw$

Let $u = w - \ln w \Rightarrow du = \left(1 - \frac{1}{w}\right) dw$:

$$\int \cos(u) du = \sin(u) + C = \sin(w - \ln w) + C$$

4. Computing the Laplace Transform from the Definition

4.1. Example 2.1 — Laplace Transform of $Ae^{-at}u(t)$

Problem: Find the Laplace transform of $f(t) = Ae^{-at}u(t)$.

Solution:

Apply the definition directly:

$$\begin{aligned} F(s) &= \int_{0^-}^{\infty} f(t) e^{-st} dt \\ &= \int_{0^-}^{\infty} A e^{-at} \cdot e^{-st} dt \\ &= A \int_0^{\infty} e^{-(s+a)t} dt \end{aligned}$$

Evaluating the integral (since e^{-ct} integrates to $-\frac{1}{c}e^{-ct}$):

$$F(s) = A \left[\frac{-1}{s+a} e^{-(s+a)t} \right]_{t=0}^{\infty}$$

Applying limits: as $t \rightarrow \infty$, $e^{-(s+a)t} \rightarrow 0$ (provided $\text{Re}(s) > -a$); at $t = 0$, $e^0 = 1$:

$$F(s) = A \left[0 - \frac{-1}{s+a} \right] = \frac{A}{s+a}$$

$$\mathcal{L}\{Ae^{-at}u(t)\} = \frac{A}{s+a}$$

4.2. Computing $\mathcal{L}\{e^{at}\}$

Apply the definition:

$$\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{(a-s)t} dt$$

Using the result $\int_0^{\infty} e^{ct} dt = \frac{-1}{c}$ provided $c < 0$ (i.e. $a - s < 0 \Rightarrow s > a$):

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \quad \text{provided } s > a$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

5. Properties of the Laplace Transform

5.1. Linearity

Linearity Property: Given $f(t)$ and $g(t)$, for any constants a and b :

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$$

Worked Example: Find $F(s)$ for $f(t) = 6e^{-5t} + e^{3t} + 5t^3 - 9$

Using linearity and the standard table:

$$\begin{aligned} F(s) &= 6 \cdot \frac{1}{s - (-5)} + \frac{1}{s - 3} + 5 \cdot \frac{3!}{s^{3+1}} - 9 \cdot \frac{1}{s} \\ &= \frac{6}{s + 5} + \frac{1}{s - 3} + \frac{30}{s^4} - \frac{9}{s} \end{aligned}$$

5.2. Differentiation in Time

For the first and second derivatives:

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

General n -th order:

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$$

5.3. Multiplication by t (Frequency Differentiation)

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}$$

This is entry **#30** in the standard table and is used when $f(t)$ contains a factor of t^n .

5.4. Integration in Time

$$\mathcal{L}\left\{\int_0^t f(v) dv\right\} = \frac{F(s)}{s}$$

5.5. Frequency Shift (First Shift Theorem)

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

6. Table of Laplace Transforms

The following table lists the 37 standard Laplace transform pairs used throughout this course. $u(t)$ denotes the unit step function and $\delta(t)$ the Dirac delta (impulse) function.

#	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1	1	$\frac{1}{s}$
2	e^{at}	$\frac{1}{s - a}$
3	$t^n, \quad n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
4	$t^p, \quad p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$
6	$t^{n-\frac{1}{2}}, \quad n = 1, 2, 3, \dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7	$\sin(at)$	$\frac{a}{s^2 + a^2}$
8	$\cos(at)$	$\frac{s}{s^2 + a^2}$
9	$t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$
10	$t \cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11	$\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$
12	$\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13	$\cos(at) - at \sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$
14	$\cos(at) + at \sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$
15	$\sin(at + b)$	$\frac{s \sin(b) + a \cos(b)}{s^2 + a^2}$
16	$\cos(at + b)$	$\frac{s \cos(b) - a \sin(b)}{s^2 + a^2}$
17	$\sinh(at)$	$\frac{a}{s^2 - a^2}$
18	$\cosh(at)$	$\frac{s}{s^2 - a^2}$
19	$e^{at} \sin(bt)$	$\frac{b}{(s - a)^2 + b^2}$

#	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
20	$e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2 + b^2}$
21	$e^{at} \sinh(bt)$	$\frac{b}{(s-a)^2 - b^2}$
22	$e^{at} \cosh(bt)$	$\frac{s-a}{(s-a)^2 - b^2}$
23	$t^n e^{at}, \quad n = 1, 2, 3, \dots$	$\frac{n!}{(s-a)^{n+1}}$
24	$f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25	$u_c(t) = u(t-c)$ <i>Heaviside Function</i>	$\frac{e^{-cs}}{s}$
26	$\delta(t-c)$ <i>Dirac Delta Function</i>	e^{-cs}
27	$u_c(t) f(t-c)$	$e^{-cs} F(s)$
28	$u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t+c)\}$
29	$e^{ct} f(t)$	$F(s-c)$
30	$t^n f(t), \quad n = 1, 2, 3, \dots$	$(-1)^n F^{(n)}(s)$
31	$\frac{1}{t} f(t)$	$\int_s^\infty F(u) du$
32	$\int_0^t f(v) dv$	$\frac{F(s)}{s}$
33	$\int_0^t f(t-\tau) g(\tau) d\tau$	$F(s) G(s)$
34	$f(t+T) = f(t)$	$\frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$
35	$f'(t)$	$sF(s) - f(0)$
36	$f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
37	$f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$

7. Worked Examples

7.1. Using Linearity Directly from the Table

Find $G(s)$ for: $g(t) = 4 \cos(4t) - 9 \sin(4t) + 2 \cos(10t)$

Using entries #7 and #8:

$$\begin{aligned} G(s) &= 4 \cdot \frac{s}{s^2 + 4^2} - 9 \cdot \frac{4}{s^2 + 4^2} + 2 \cdot \frac{s}{s^2 + 10^2} \\ &= \frac{4s}{s^2 + 16} - \frac{36}{s^2 + 16} + \frac{2s}{s^2 + 100} \end{aligned}$$

Find $H(s)$ for: $h(t) = 3 \sinh(2t) + 3 \sin(2t)$

Using entries #17 and #7:

$$\begin{aligned} H(s) &= 3 \cdot \frac{2}{s^2 - 2^2} + 3 \cdot \frac{2}{s^2 + 2^2} \\ &= \frac{6}{s^2 - 4} + \frac{6}{s^2 + 4} \end{aligned}$$

Find $G(s)$ for: $g(t) = e^{3t} + \cos(6t) - e^{3t} \cos(6t)$

Using entries #2, #8, and #20:

$$\begin{aligned} G(s) &= \frac{1}{s - 3} + \frac{s}{s^2 + 6^2} - \frac{s - 3}{(s - 3)^2 + 6^2} \\ &= \frac{1}{s - 3} + \frac{s}{s^2 + 36} - \frac{s - 3}{(s - 3)^2 + 36} \end{aligned}$$

7.2. Using Entry #30: Multiplication by t^n

Entry #30 states: $\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s)$

Find $F(s)$ for: $f(t) = t \cosh(3t)$

This function is not directly in the table. Use entry #30 with $n = 1$:

$$F(s) = \mathcal{L}\{t \cdot g(t)\} = -G'(s) \quad \text{where } g(t) = \cosh(3t)$$

From entry #18:

$$G(s) = \frac{s}{s^2 - 9}$$

Differentiating:

$$G'(s) = -\frac{s^2 + 9}{(s^2 - 9)^2}$$

Therefore:

$$F(s) = -G'(s) = \frac{s^2 + 9}{(s^2 - 9)^2}$$

Find $H(s)$ for: $h(t) = t^2 \sin(2t)$

Use entry #30. We can apply it in two ways:

Method 1 — use $n = 1$ twice (i.e. with $f(t) = t \sin(2t)$):

$$H(s) = \mathcal{L}\{t \cdot f(t)\} = -F'(s) \quad \text{where } f(t) = t \sin(2t)$$

From entry #9: $F(s) = \frac{4s}{(s^2 + 4)^2}$

$$F'(s) = -\frac{12s^2 - 16}{(s^2 + 4)^3}$$

Therefore:

$$H(s) = -F'(s) = \frac{12s^2 - 16}{(s^2 + 4)^3}$$

Method 2 — use $n = 2$ directly (i.e. with $f(t) = \sin(2t)$):

$$H(s) = (-1)^2 F''(s) = F''(s) \quad \text{where } F(s) = \frac{2}{s^2 + 4}$$

Both methods give the same result.

7.3. Using Entry #24: Scaling in Time

Entry #24 states: $\mathcal{L}\{f(ct)\} = \frac{1}{c} F\left(\frac{s}{c}\right)$

Find $G(s)$ for: $g(t) = t^{3/2}$

Use entry #32 (integration) along with entry #5 ($\sqrt{t}$). Noting:

$$\int_0^t \sqrt{v} dv = \frac{2}{3} t^{3/2} \quad \Rightarrow \quad t^{3/2} = \frac{3}{2} \int_0^t \sqrt{v} dv$$

From entry #5: $f(t) = \sqrt{t} \Rightarrow F(s) = \frac{\sqrt{\pi}}{2s^{3/2}}$

Applying entry #32:

$$G(s) = \frac{3}{2} \cdot \frac{F(s)}{s} = \frac{3}{2} \cdot \frac{\sqrt{\pi}}{2s^{3/2}} \cdot \frac{1}{s} = \frac{3\sqrt{\pi}}{4s^{5/2}}$$

Find $F(s)$ for: $f(t) = (10t)^{3/2}$

Note that $f(t) = g(10t)$ where $g(t) = t^{3/2}$ from Example F.

Applying entry #24 with $c = 10$:

$$F(s) = \frac{1}{10} G\left(\frac{s}{10}\right) = \frac{1}{10} \cdot \frac{3\sqrt{\pi}}{4 \left(\frac{s}{10}\right)^{5/2}}$$

Simplifying:

$$F(s) = \frac{1}{10} \cdot \frac{3\sqrt{\pi} \cdot 10^{5/2}}{4s^{5/2}} = \frac{10^{3/2} \cdot 3\sqrt{\pi}}{4s^{5/2}}$$

8. Key Formulas Summary

Name	Formula
Laplace definition	$F(s) = \int_{0^-}^{\infty} f(t) e^{-st} dt$
Linearity	$\mathcal{L}\{af + bg\} = aF(s) + bG(s)$
Exponential	$\mathcal{L}\{e^{at}\} = \frac{1}{s - a}$
Exponential (damped)	$\mathcal{L}\{Ae^{-at}u(t)\} = \frac{A}{s + a}$
First derivative	$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$
Second derivative	$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$
Multiplication by t	$\mathcal{L}\{tf(t)\} = -F'(s)$
Integration	$\mathcal{L}\left\{\int_0^t f dv\right\} = \frac{F(s)}{s}$
Frequency shift	$\mathcal{L}\{e^{ct}f(t)\} = F(s - c)$
Time scaling	$\mathcal{L}\{f(ct)\} = \frac{1}{c}F\left(\frac{s}{c}\right)$

Glossary of Key Terms

Term	Definition
Laplace Transform	A mathematical operation that converts a time-domain function $f(t)$ into a complex frequency-domain function $F(s)$, defined as $F(s) = \int_0^\infty f(t)e^{-st}dt$.
Complex Variable s	The variable $s = \sigma + j\omega$, where σ is the real part and ω is the imaginary part. It represents complex frequency.
Inverse Laplace Transform	The operation $\mathcal{L}^{-1}\{F(s)\} = f(t)$ that recovers the time-domain function from its Laplace transform.
Transfer Function	The Laplace-domain ratio of output to input for a linear, time-invariant system at rest: $G(s) = Y(s)/U(s)$.
Frequency Domain	The representation of signals and systems in terms of the complex variable s rather than time t .
Time Domain	The standard representation of signals as functions of time t .
Linearity Property	$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$. The Laplace transform of a linear combination equals the linear combination of transforms.
Differentiation Property	$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$. Differentiation in time corresponds to multiplication by s in the s -domain.
Integration Property	$\mathcal{L}\left\{\int_0^t f(v)dv\right\} = F(s)/s$. Integration in time corresponds to division by s .
Frequency Shift	$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$. Multiplying by an exponential in time shifts the transform in s .
Time Scaling	$\mathcal{L}\{f(ct)\} = \frac{1}{c}F(s/c)$. Compressing time expands the s -domain and vice versa.
Unit Step Function	$u(t) = 0$ for $t < 0$ and $u(t) = 1$ for $t \geq 0$. Its Laplace transform is $1/s$.
Heaviside Function	$u_c(t) = u(t - c)$, a unit step delayed by c . Laplace transform: e^{-cs}/s .
Dirac Delta (Impulse)	$\delta(t - c)$, an ideal impulse of unit area at $t = c$. Laplace transform: e^{-cs} .
Convolution	$\int_0^t f(t - \tau)g(\tau) d\tau \leftrightarrow F(s)G(s)$. Convolution in time equals multiplication in s .
Partial Fractions	A method for decomposing a rational $F(s)$ into simpler terms whose inverse transforms are known.
Region of Convergence	The set of s values for which the Laplace integral converges (e.g. $\text{Re}(s) > a$ for e^{at}).
Chain Rule	$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$. Used when differentiating composite functions.
Product Rule	$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$. Used when two functions are multiplied.
Substitution Rule	$\int f(g(x))g'(x)dx = \int f(u)du$ where $u = g(x)$. Simplifies integrals with composite functions.

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